INTRODUCTION

Partial Fractions (Algebra)

$\frac{A}{ax+b}$ for a linear factor $(ax+b)$

$\frac{A}{ax+b} + \frac{B}{(ax+b)^2}$ for repeated linear factors

$\frac{Ax+B}{ax^2+bx+c}$ for a quadratic factor (ax^2+bx+c)

Logarithms and Exponentials

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$e^x = \cosh x + \sinh x$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$\tan x = \frac{e^{ix} - e^{-ix}}{i(e^{ix} + e^{-ix})}$$

$$e^{ix} = \cos x + i \sin x$$

$$\cos^2 x + \sin^2 x = 1$$

$$\log_a N = x \Rightarrow a^x = N$$

$$\log_a N = \frac{\log_b N}{\log_b a}, \qquad \log AB = \log A + \log B$$

$$\log \frac{A}{B} = \log A - \log B, \qquad \log A^m = m \log A$$

Example: Prove that $\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)$

$$y = \sinh^{-1} x \Rightarrow x = \sinh y \Rightarrow \ln\left(x + \sqrt{x^2+1}\right)$$

$$= \ln\left(\sinh y + \sqrt{\sinh^2 y + 1}\right) = \ln\left(\sinh y + \cosh y\right)$$

$\cosh^2 y - \sinh^2 y = 1$

$$= \ln\left(e^y\right) = y = \sinh^{-1} x$$

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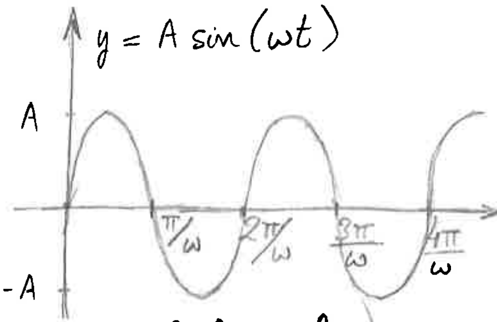
Trigonometry

amplitude, phase angle

$$y(t) = A \sin(\omega t + \alpha)$$ Eq. of a wave

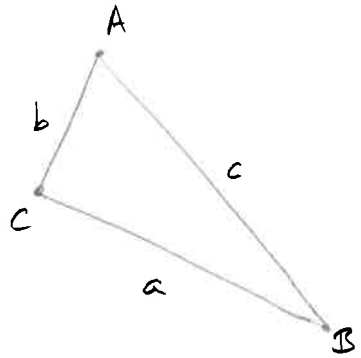
angular frequency



period: $T = \frac{2\pi}{\omega}$ (time to complete one full cycle)

frequency: $f = \frac{1}{T} = \frac{\omega}{2\pi}$ (number of cycles completed in 1 second)

Sine rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\cos C}$



Cosine rule:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

DIFFERENTIATION

$$\text{Rate of change of } y(x) \text{ (for any } x) = \lim_{\delta x \to 0} \frac{y(x+\delta x) - y(x)}{\delta x} = \frac{dy}{dx}$$

<u>Example</u>: Show that $\frac{d}{dx}\Big(a f(x) + b g(x)\Big) = a \frac{df}{dx} + b \frac{dg}{dx}$

$$\frac{d}{dx}\Big(a f(x) + b g(x)\Big) = \lim_{\delta x \to 0} \frac{a f(x+\delta x) + b g(x+\delta x) - \big(a f(x) + b g(x)\big)}{\delta x}$$

$$= \lim_{\delta x \to 0} \frac{a\big[f(x+\delta x) - f(x)\big]}{\delta x} + \lim_{\delta x \to 0} \frac{b\big[g(x+\delta x) - g(x)\big]}{\delta x}$$

$$= a \frac{df}{dx} + b \frac{dg}{dx} \quad \#$$

<u>Rules</u>:

$$y(x) = u(x) v(x) \Rightarrow \frac{dy}{dx} = u'v + uv'$$

$$y(x) = \frac{u(x)}{v(x)} \Rightarrow \frac{dy}{dx} = \frac{u'v - uv'}{v^2}$$

$$y = y(z(x)) \Rightarrow \frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx}$$

$$y = \ln(f(x)) \Rightarrow \frac{dy}{dx} = \frac{1}{f(x)} \frac{df}{dx}$$

$$x(t), y(t) \Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \Big/ \frac{dx}{dt}$$

Implicit functions cannot be written in the form $y = f(x)$, but are given by $f(x, y(x)) = 0$

Example: Find $\frac{dy}{dx}$ given $y^2 + x = x^3 + 3y$, with $y(x)$

$$2y\frac{dy}{dx} + 1 = 3x^2 + 3\frac{dy}{dx} \implies \frac{dy}{dx} = \frac{3x^2 - 1}{2y - 3}$$

$$\frac{d(y^2)}{dx} = \frac{dz}{dy}\frac{dy}{dx} = 2y\frac{dy}{dx}$$

Chain rule with $z = y^2$

Logarithmic Differentiation

Example: Find $\frac{df}{dx}$ given $f(x) = \frac{e^x x^3 \ln x}{(1+e^{3x})\sin(x^2)}$

$$f = \frac{u\,v\,w}{\alpha\beta}$$

$u = e^x$, $\quad v = x^3$, $\quad w = \ln x$, $\quad \alpha = 1 + e^{3x}$, $\quad \beta = \sin x^2$

$u' = e^x$, $\quad v' = 3x^2$, $\quad w' = \frac{1}{x}$, $\quad \alpha' = 3e^{3x}$, $\quad \beta' = 2x\cos x^2$

$$\ln f = \ln u + \ln v + \ln w - \ln\alpha - \ln\beta$$

$$\frac{1}{f}\frac{df}{dx} = \frac{u'}{u} + \frac{v'}{v} + \frac{w'}{w} - \frac{\alpha'}{\alpha} - \frac{\beta'}{\beta}$$

$$\implies \frac{df}{dx} = \frac{e^x x^3 \ln x}{(1+e^{3x})\sin x^2}\left[1 + \frac{3}{x} + \frac{1}{x\ln x} - \frac{3}{1+e^{-3x}} - 2x\cot x^2\right]$$

Tangents and Normals

$t = m_1 x + c_1$

$m_1 \cdot m_2 = -1$

$n = m_2 x + c_2$

Maximum and Minimum Values of a Function

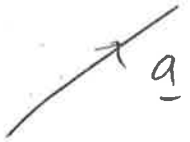
$\frac{dy}{dx} = 0$ or $\frac{dy}{dx}$ doesn't exist

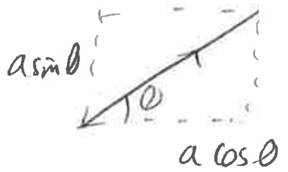
$\frac{d^2y}{dx^2} < 0 \Rightarrow$ Maximum

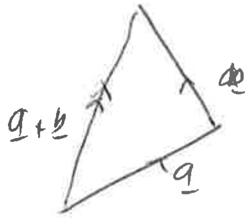
$\frac{d^2y}{dx^2} > 0 \Rightarrow$ Minimum

$\frac{d^2y}{dx^2} = 0 \Rightarrow$ Point of inflection

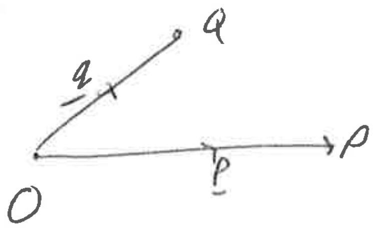
Vectors







$$\hat{\underline{a}} = \frac{\underline{a}}{a}$$



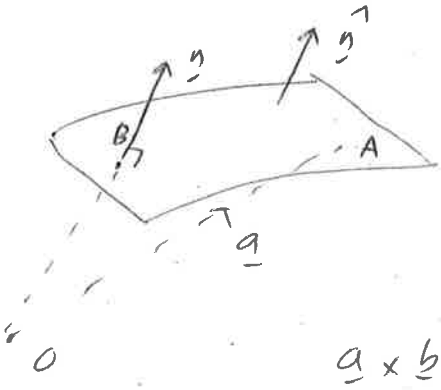
$$PQ = \underline{q} - \underline{p}$$

$$\underline{r} = a\underline{i} + b\underline{j} = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\underline{a}\cdot\underline{b} = a_1 b_1 + a_2 b_2$$

$$\underline{a}\cdot\underline{b} = ab\cos\theta$$

$$\underline{i}\cdot\underline{i} = 1 \quad , \quad \underline{i}\cdot\underline{j} = 0 \ldots$$



$$\underline{a}\times\underline{b} = ab\sin\theta\,\hat{\underline{e}}$$

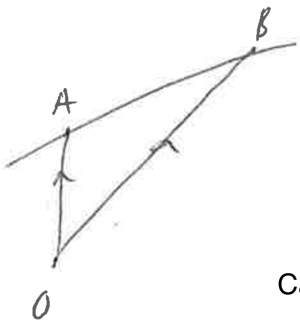
$$\underline{a}\times\underline{b} = -\underline{b}\times\underline{a}$$

$$\underline{i}\times\underline{i} = 0$$

$$\underline{i}\times\underline{j} = \underline{k} \ldots$$

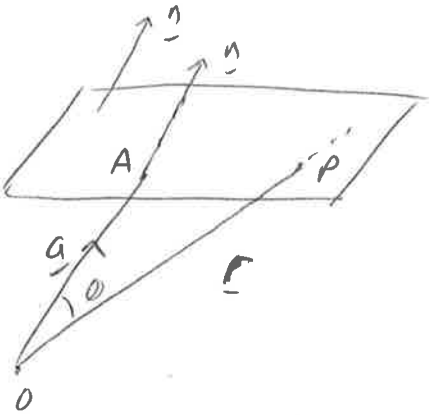
$$\underline{a}\times\underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2 b_3 - a_3 b_2)\underline{i}$$

$$-\underline{j}(a_1 b_3 - a_3 b_1) + \underline{k}(a_1 b_2 - a_2 b_1)$$



Vector form: $\underline{r} = \underline{a} + \lambda(\underline{b} - \underline{a})$

Cartesian form: $\dfrac{x - a_1}{b_1 - a_1} = \dfrac{y - a_2}{b_2 - a_2} = \dfrac{z - a_3}{b_3 - a_3}$



$\overrightarrow{AP} = \underline{r} - \underline{a}$

$(\underline{r} - \underline{a}) \cdot \underline{n} = 0$

$\underline{r} \cdot \hat{\underline{n}} = d$ if n is a unit vector

$(\underline{r} - \underline{a}) \cdot \underline{n} = 0 \Rightarrow \underline{r} \cdot \underline{n} = \underline{a} \cdot \underline{n}$

Vector form

Note that the Cartesian form of these equations is obtained by letting r = x i + y j + z k and evaluating the scalar product.

Matrices

$(A)_{ij} = a_{ij}$ $\qquad$ $(A^T)_{ij} = (A)_{ji}$

$\mathrm{Tr}(A) = \sum_{i=1}^{n} A_{ii}$ $\qquad$ $\lambda(A)_{ij} = \lambda a_{ij}$

Add square matrices

$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$

$$\underset{m \times n}{A} \times \underset{p \times q}{B} = \underset{m \times q}{C}$$

$n = p$

$$\det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

minor of $a_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

cofactor $=$ minor $\times (-1)^{i+j}$ $\qquad$ $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$

$\det(AB) = \det(A)\det(B)$

$$A^{-1} = \frac{1}{|A|}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \qquad A^{-1} = \frac{\mathrm{adj}(A)}{|A|}$$

$A^{+} \rightarrow$ replace each element by cofactor.

Transformation $T\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}x'\\y'\end{pmatrix}$

scale $\begin{pmatrix}\alpha & 0\\0 & \beta\end{pmatrix}$

rot $\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$

refl. (x) $\begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}$

Translate $\begin{pmatrix}x\\y\end{pmatrix} \rightarrow \begin{pmatrix}x\\y\\1\end{pmatrix} \Rightarrow T = \begin{pmatrix}1 & 0 & t_x\\0 & 1 & t_y\\0 & 0 & 1\end{pmatrix} \rightarrow \begin{pmatrix}x+t_x\\y+t_y\end{pmatrix}$

$$AX = \lambda X$$

$$(A - \lambda I)X = 0 \Rightarrow \det(A - \lambda I) = 0$$

$$A\cancel{B} = \cancel{B}$$

$$X = A^{-1}B$$

INTEGRATION

Differentiation in reverse: $\int f(x)\,dx = F(x) + C$, $\frac{dF(x)}{dx} = f(x)$ (antiderivative: $F(x)$)

Fundamental Theorem of Calculus: $\lim_{\delta x \to 0} \sum_{x=a}^{x=b} f(x)\,\delta x = \int_a^b f(x)\,dx = F(b) - F(a)$

Rules:

$$\int_a^b \left(f(x) \pm g(x)\right) dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

$$\int_a^b k f(x)\,dx = k \int_a^b f(x)\,dx$$

$$\int_a^b u\left(\frac{dv}{dx}\right) dx = uv\Big]_a^b - \int_a^b v\left(\frac{du}{dx}\right) dx$$

$$\int_{g(a)}^{g(b)} f(x)\,dx = \int_a^b f(g(x))\,\frac{dg}{dx}\,dx$$

Sometimes it helps using partial fractions for integrals of algebraic fractions and using trigonometric identities for integrals of trigonometric functions.

Example:

$$\int 3x^5 e^{x^3} dx = \int x^3\, 3x^2 e^{x^3} dx = x^3 e^{x^3} - \int 3x^2 e^{x^3} dx$$

$u = x^3$, $u' = 3x^2$
$v' = 3x^2 e^{x^3}$, $v = e^{x^3}$

$$= x^3 e^{x^3} - e^{x^3} = e^{x^3}(x^3 - 1) + C$$